Development of Finite Element Solver for Light Scattering Simulation

Lee Shien Wei

Faculty of Engineering
UNiversiti Malaya
2026
Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Engineering (Hons) Mechanical Engineering

Declaration of Original Literary Work

I, **Lee Shien Wei**, declare that this thesis entitled *"Development of Finite Element Solver for Light Scattering Simulation"* is the result of my own work except as cited in the references. The thesis has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

Lee Shien Wei
November 24, 2025

Abstract

Light scattering simulation is a fundamental component in optical engineering and physics. This project aims to develop a custom Finite Element Method (FEM) solver to address the limitations of current commercial software. By utilizing the FEniCSx framework, we implement a solver for the Helmholtz equation...

Acknowledgements

I would like to express my sincere gratitude to my supervisor for his guidance. His expertise and insights were invaluable throughout this project. I also wish to thank my family and friends for their unwavering support and encouragement.

Finally, I would like to acknowledge the resources and tools that facilitated this research, including the FEniCSx framework and the academic community that contributed to its development.

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Chapter 1

Introduction

1.1 Background

Provide the context of light scattering.

1.2 Literature Review

Discuss existing solutions and their limitations.

1.3 Problem Statement

Current commercial solvers are ’black boxes’.

Anees and Angermann [2019] provides an open-source alternative. Van and Wood [2004] discuss FEM for light scattering.

1.4 Objectives

- To derive the weak variational formulation.
- To implement a Finite Element solver using FEniCSx.
- To validate the solver against analytical Mie scattering solutions.

Chapter 2

Mathematical Formulation of Light Scattering Problem

2.1 Derivation of Maxwell's Equations

Start from the fundamental Maxwell's equations:

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \tag{2.1}$$

$$\nabla \times \boldsymbol{H} = \boldsymbol{J} + \frac{\partial \boldsymbol{D}}{\partial t} \tag{2.2}$$

Derive the vector wave equation (Helmholtz equation) for the electric field $\boldsymbol{E}$.

2.2 Boundary Conditions

Define the interface conditions between different media (e.g., scatterer and background).

$$\boldsymbol{n} \times (\boldsymbol{E}_1 - \boldsymbol{E}_2) = 0 \tag{2.3}$$

2.3 Initial-Boundary Conditions

Discuss the radiation condition (Sommerfeld radiation condition) required to simulate an infinite domain without reflection artifacts.

$$\lim_{r \to \infty} r \left(\nabla \times \boldsymbol{E} - ik\boldsymbol{n} \times \boldsymbol{E} \right) = 0 \tag{2.4}$$

2.4 Variational Formulation of Light Scattering Problem

This is the core section for FEM. Multiply the strong form by a test function $\boldsymbol{v}$ and integrate by parts to obtain the weak form used in FEniCSx.

Chapter 3

Numerical Formulation of Light Scattering Problem

3.1 Galerkin Approximations

Explain how the infinite-dimensional function space is approximated.

Chapter 4

Implementation of Numerical Method

4.1 FEniCSx Software

4.2 Implementation Overview

4.3 UFL problem description

4.4 The FEniCSX Light Scattering Solver

Chapter 5

Validation of finite element solver

5.1 Parameters and Configuration

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Chapter 6

Conclusion

References

Asad Anees and Lutz Angermann. Time domain finite element method for maxwell's equations. *IEEE Access*, 7:63852–63867, 2019. doi: 10.1109/ACCESS.2019.2916394.

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